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ON

Empirical Analysis of Virtual Connectivity and Decision Fatigue Among Corporate Managers

Submitted in partial fulfilment of the requirements for the award of degree of

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work presented in this project report entitled "Empirical Analysis of Virtual Connectivity and Decision Fatigue Among Corporate Managers" is an authentic record of our own work carried out under the supervision of Preeti Saini, Assistant Professor, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India. The matter presented in this project report has not been submitted in any other university or institute for the award of any degree.

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ABSTRACT

In today's fast-paced digital environment, managers are continuously exposed to information flow, multitasking, and after-hours engagement through emails, instant messaging platforms, and collaborative tools. While digital technologies improve responsiveness and efficiency, they also impose cognitive costs that may affect judgment and decision quality. This project investigates the relationship between digital connectivity intensity and decision fatigue among corporate managers, and further examines how cognitive exhaustion influences perceived decision quality. A mixed-methods approach was adopted, combining a structured survey with 200 valid responses from middle and senior managers across information technology, consulting, finance, and manufacturing sectors, together with semi-structured interviews. The quantitative analysis, performed using Python-based tools, indicates that digital connectivity intensity and after-hours engagement are positively associated with decision fatigue, while decision fatigue is negatively associated with perceived decision quality. The findings support the need for structured digital boundaries and recovery-oriented management practices to sustain cognitive performance in digitally intensive workplaces.

Keywords— Decision Fatigue, Digital Connectivity, Cognitive Exhaustion, Managerial Decision-Making, Decision Quality, Cognitive Sustainability, Regression Analysis

1. INTRODUCTION

1.1 Background and Relevance

The widespread adoption of digital communication technologies has transformed managerial work into a continuously connected process. Managers are expected to respond to emails, messages, approvals, and team coordination tasks throughout the day, often with minimal cognitive recovery. Although these tools improve communication efficiency and operational responsiveness, they also increase interruptions, task-switching, and mental load, thereby creating conditions that are conducive to decision fatigue.

1.2 Problem Statement

Despite the growing attention given to technostress, burnout, and information overload, comparatively less research has examined the specific relationship between digital connectivity intensity and decision fatigue in managerial contexts. The problem addressed in this study is whether continuous engagement with digital platforms contributes to cognitive exhaustion and whether that exhaustion affects perceived decision quality.

1.3 Objectives

The project pursues two main objectives: first, to examine the relationship between digital connectivity and decision fatigue among managers; and second, to assess how cognitive exhaustion influences perceived decision quality. In addition, the study explores practical mechanisms, such as digital boundaries and recovery practices, that may help reduce fatigue in digitally intensive workplaces.

2. METHODOLOGY

2.1 Research Design

The research adopted a mixed-methods design combining quantitative survey analysis with qualitative interview insights. Three primary constructs were studied: digital connectivity intensity, decision fatigue, and perceived decision quality. The quantitative component enabled statistical testing of relationships among variables, while the qualitative component helped interpret the lived experiences of managers working in digitally connected environments.

2.2 Data Sources and Sampling

Data were collected from 200 valid respondents drawn from middle- and senior-level managers across digitally integrated industries, including information technology, consulting, finance, and manufacturing. A purposive sampling strategy was used to ensure that the participants had regular exposure to digital communication tools such as email, Microsoft Teams, Slack, and related platforms. The sample reflected a diverse managerial cohort with high exposure to continuous connectivity and extended work hours.

2.3 Research Instrument

A structured questionnaire using a five-point Likert scale was employed for data collection. The instrument consisted of sections covering demographic details, digital connectivity patterns, decision fatigue, perceived decision quality, and coping practices. Reliability was assessed through internal consistency, with Cronbach's alpha used as the primary criterion for measurement stability.

2.4 Analytical Approach

The quantitative data were analyzed in Python using pandas, statsmodels, matplotlib, and seaborn. Descriptive statistics were used to summarize the sample, Pearson correlation analysis was used to estimate pairwise relationships, and multiple regression was used to assess the predictive impact of digital connectivity and after-hours engagement on decision fatigue. Open-ended responses were analyzed using thematic analysis to identify common patterns such as cognitive overload, boundary blurring, and adaptive coping.

Table I. Comparison of Existing Studies

Study	Focus Area	Limitation
Simon (1947)	Bounded rationality	No digital context
Baumeister (1998)	Ego depletion	Not managerial
Danziger (2011)	Decision fatigue	Non-digital
Tarafdar et al.	Technostress	No decision quality focus

Marks (2014)	Interruptions	Short-term
Sarker (2019)	Connectivity	Limited empirical validation

The comparison shows that prior studies have examined bounded rationality, ego depletion, technostress, interruptions, and always-on connectivity, but few have integrated these perspectives into a managerial model that links digital connectivity directly to decision fatigue and decision quality.

2.5 Data Handling and Processing

The dataset was exported from Google Forms in CSV format and processed using Python. Data cleaning involved the removal of incomplete or inconsistent responses, while data transformation converted categorical responses into numerical values suitable for analysis. Feature construction was used to aggregate responses into scores for connectivity, fatigue, and decision quality. After preprocessing, the dataset was analyzed using descriptive statistics, correlation analysis, and regression modeling.

2.6 Regression Models

The relationships among the main constructs were modeled using the following regression equations:

$$Decision\ Fatigue = \beta_0 + \beta_1(Connectivity) + \beta_2(AfterHours) + \varepsilon$$

$$Decision\ Quality = \beta_0 + \beta_1(Decision\ Fatigue) + \varepsilon$$

Here, β_0 denotes the intercept, β_1 and β_2 denote the regression coefficients, and ε represents the error term. These equations were used to evaluate the strength and direction of the relationships between the variables.

3. TOOLS AND TECHNOLOGIES

The project was implemented using a combination of survey tools, analytical software, and visualization libraries. Google Forms was used for data collection, while Python served as the main analysis environment. The primary libraries included pandas for data manipulation, NumPy for numerical computation, matplotlib and seaborn for data visualization, and statsmodels for correlation and regression analysis. Microsoft Word was used to prepare the report, and GitHub was designated as the repository for code submission, documentation, and version control.

Table II. Parameters Used in the Study

S.No.	Parameter	Description
1	Digital Connectivity	Frequency of interaction with digital platforms
2	Decision Fatigue	Level of cognitive exhaustion experienced
3	Decision Quality	Effectiveness and accuracy of decisions
4	After-Hours Connectivity	Extent of work-related engagement beyond official hours
5	Multitasking Level	Degree of simultaneous handling of multiple digital tasks
6	Response Time Pressure	Urgency to respond to messages or tasks
7	Working Hours	Average number of working hours per day
8	Experience Level	Years of professional or managerial experience
9	Industry Type	Sector in which the respondent is working

4. IMPLEMENTATION

4.1 Survey Design and Data Collection

The implementation began with the design of a structured questionnaire that captured the intensity of digital connectivity, the frequency of after-hours engagement, the extent of multitasking, the level of perceived mental exhaustion, and the perceived quality of decisions. The survey was distributed to managerial respondents across digitally intensive industries, and 200 valid responses were retained for final analysis.

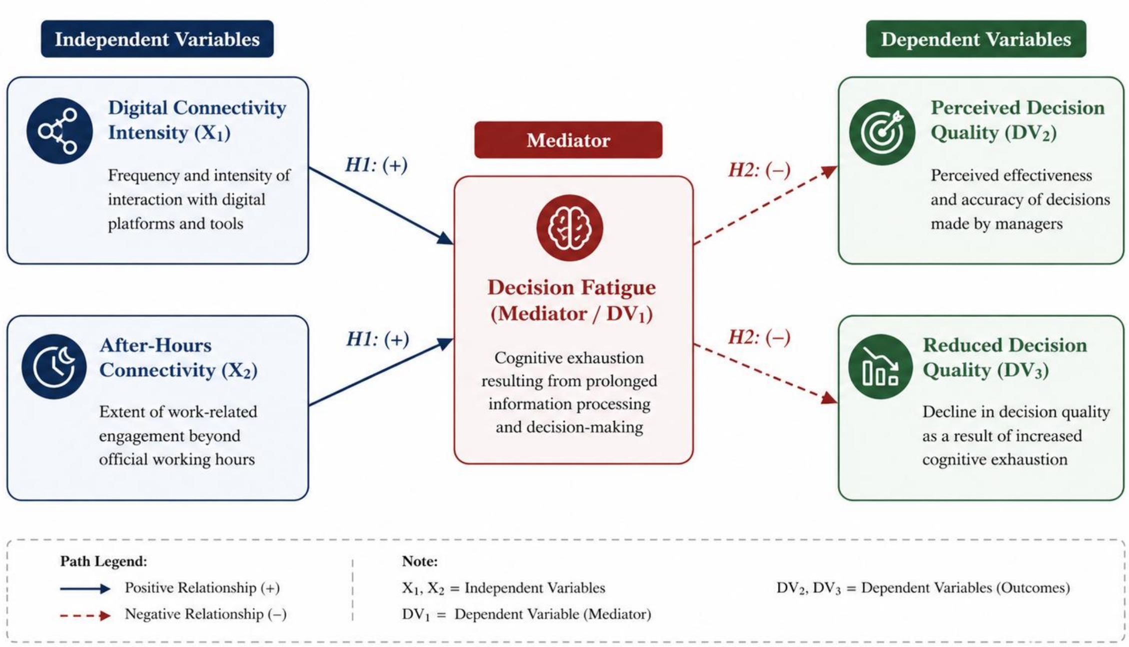
4.2 Preprocessing Workflow

After collection, the survey data were screened for completeness and consistency. Responses were normalized, categorical items were converted to numeric scores, and composite indicators were calculated for the major constructs. This preprocessing ensured that the dataset was ready for statistical testing and visualization.

4.3 Visual Analytics

Four core visualizations were prepared to interpret the study: the conceptual model, the demographic profile of respondents, the key indicators of digital strain, and the mean scores of key survey dimensions. These figures help present the structure of the study, the characteristics of the sample, and the main observed patterns.

Figure 1. Conceptual Research Model and Hypotheses



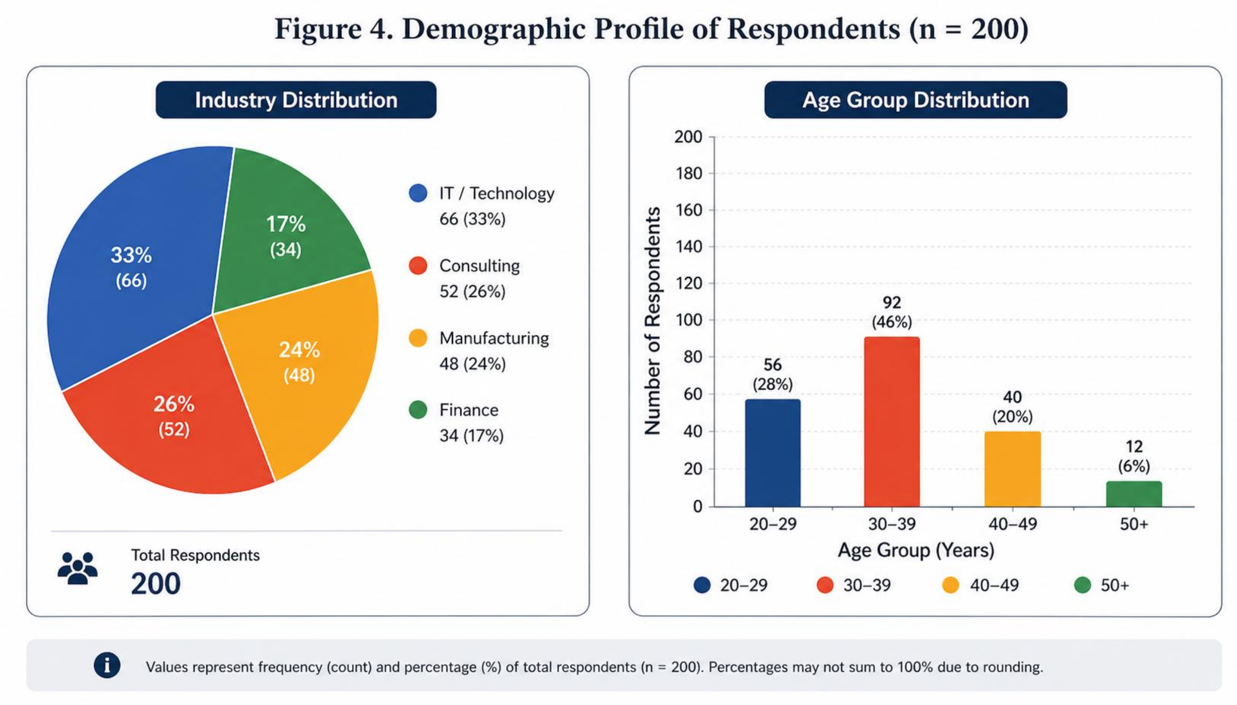


Fig. 2. Demographic Profile of Respondents (n = 200)

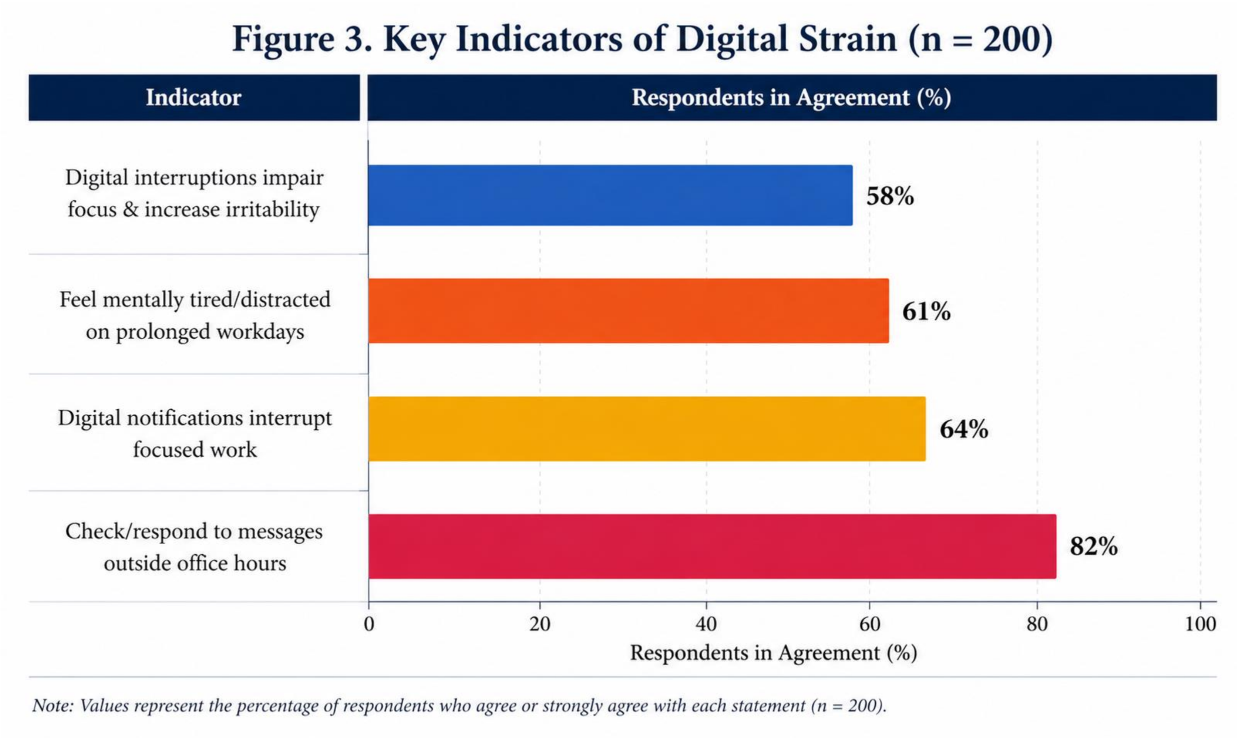


Fig. 3. Key Indicators of Digital Strain (n = 200)

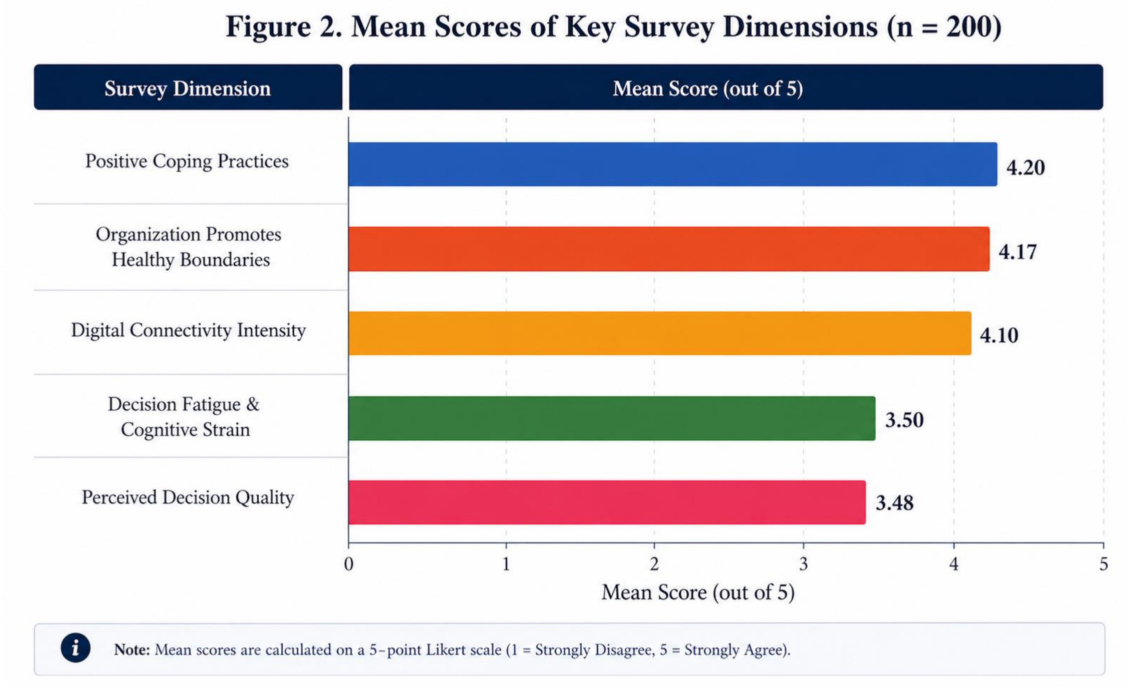
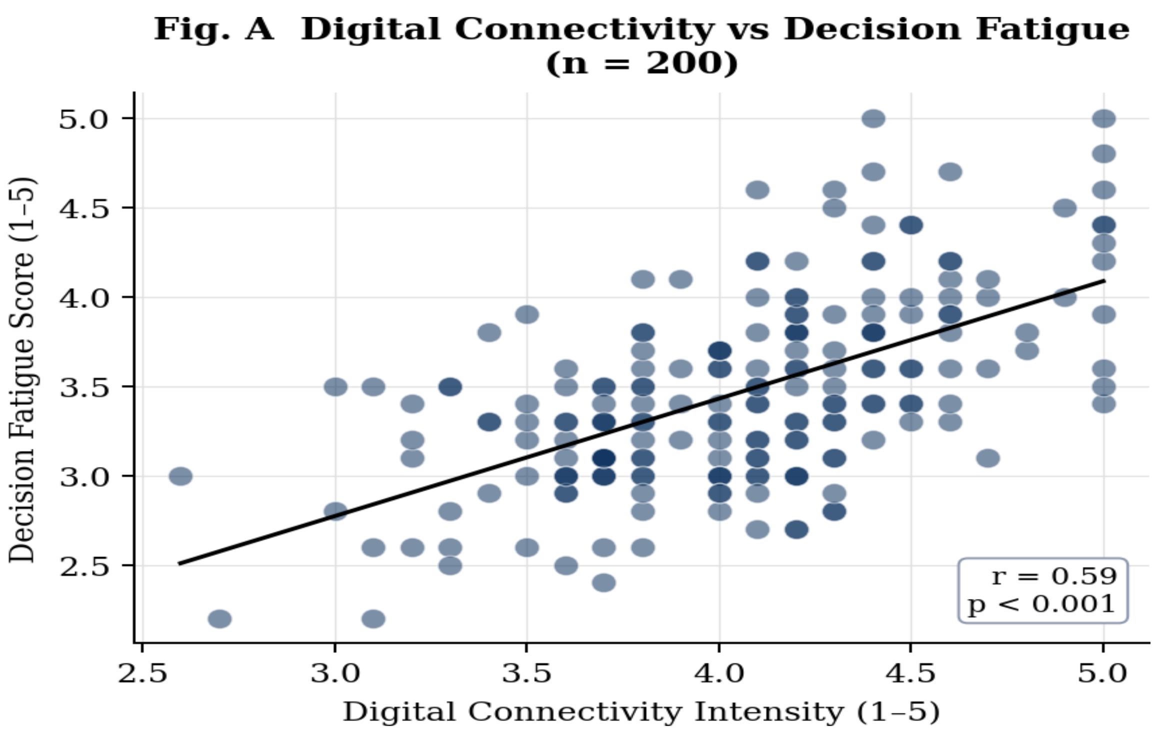
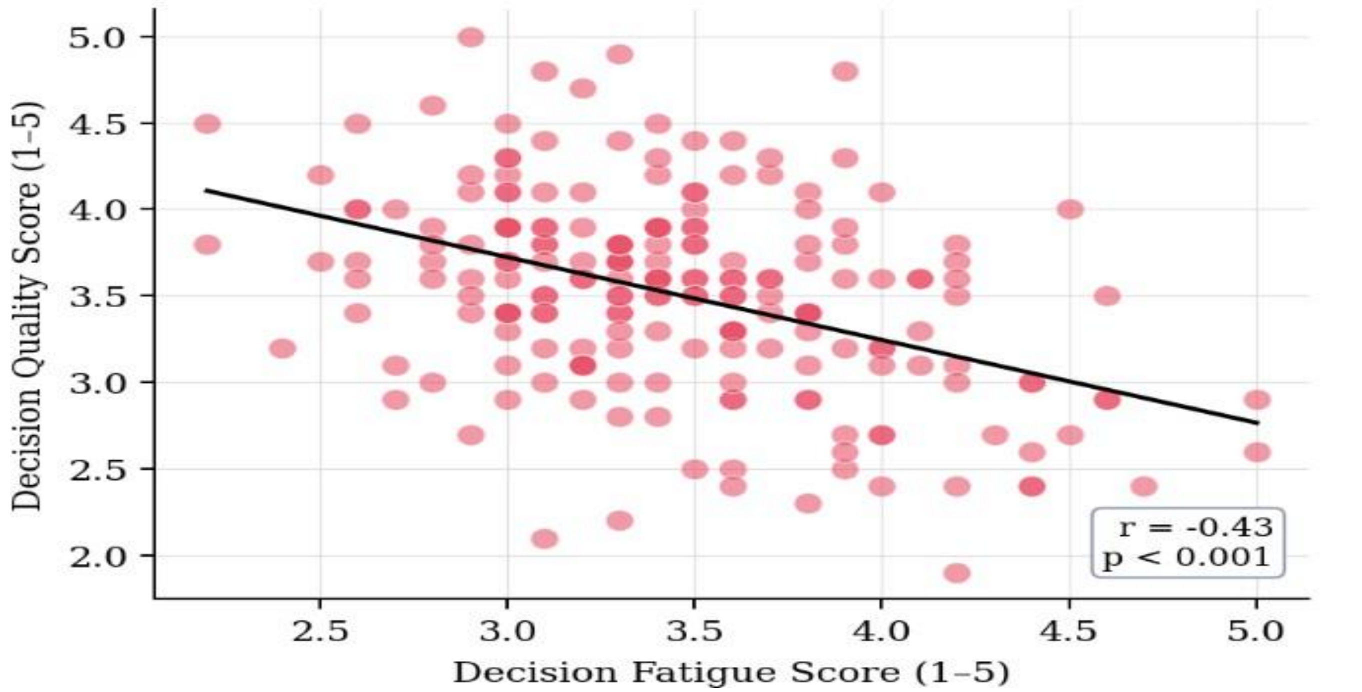


Fig. 4. Mean Scores of Key Survey Dimensions (n = 200)



Digital Connectivity Intensity vs Decision Fatigue ($r = 0.59$, $p < 0.001$, $n = 200$)

Fig. B Decision Fatigue vs Perceived Decision Quality (n = 200)



Correlation Matrix — Key Study Variables (n = 200)

5. MAJOR FINDINGS / OUTCOMES / RESULTS

5.1 Demographic Profile

The final sample consisted of 200 respondents. In the demographic profile, the largest age group was 30–39 years (46%), followed by 20–29 years (28%), 40–49 years (20%), and 50+ years (6%). In terms of industry, 33% of respondents were from IT/Technology, 26% from Consulting, 24% from Manufacturing, and 17% from Finance. The sample therefore represented managers from digitally intensive sectors.

5.2 Descriptive Results

The average score for digital connectivity intensity was 4.10 out of 5, indicating a high level of constant interaction with digital platforms. Decision fatigue had a mean of 3.50, reflecting moderate-to-high mental exhaustion, while perceived decision quality had the lowest mean at 3.48. Positive coping practices and organizational boundary support scored comparatively higher, indicating that many respondents already employ or benefit from fatigue-reducing behaviors.

5.3 Correlation and Regression Findings

The correlation analysis showed that digital connectivity intensity was positively associated with decision fatigue ($r = 0.59$, $p < 0.001$), and decision fatigue was negatively associated with perceived decision quality ($r = -0.43$, $p < 0.001$). After-hours engagement also showed a strong positive association with connectivity intensity ($r = 0.67$, $p < 0.001$) and with fatigue ($r = 0.59$, $p < 0.001$). In the regression model, digital connectivity intensity and after-hours engagement together explained approximately 41% of the variance in decision fatigue (Adjusted $R^2 = 0.41$, $p < 0.001$).

5.4 Qualitative Insights

The open-ended responses and interviews revealed three dominant themes: cognitive overload and fragmented attention, blurred personal and professional boundaries, and adaptive coping practices. Respondents frequently cited constant notifications, repeated task switching, and the expectation of immediate response as major contributors to fatigue. Organizations that discouraged after-hours email practices were perceived as more supportive and cognitively sustainable.

5.5 Key Summary Table

Dimension	Key Finding
Digital Connectivity	Mean = 4.10/5; frequent after-hours engagement observed
Decision Fatigue	Mean = 3.50/5; moderate-to-high cognitive exhaustion
Decision Quality	Mean = 3.48/5; lowest among major constructs

Connectivity ↔ Fatigue	$r = 0.59$, $p < 0.001$
Fatigue ↔ Decision Quality	$r = -0.43$, $p < 0.001$
Regression	Adj. $R^2 = 0.41$; connectivity and after-hours engagement were significant predictors

6. CONCLUSION AND FUTURE SCOPE

The project demonstrates that virtual or digital connectivity is a significant predictor of decision fatigue in managerial environments. As managers remain continuously connected through digital platforms, cognitive resources are depleted, which in turn affects decision quality. The empirical findings support the need for digital boundaries, structured recovery periods, and organizational practices that reduce unnecessary interruptions.

The study can be extended in several directions. Future work may include larger and more diverse samples, advanced modeling techniques such as structural equation modeling and machine learning, and the incorporation of real-time behavioral or biometric data. Additional research may also evaluate AI-based decision support systems that help prioritize tasks and reduce cognitive overload in digitally intensive organizations.

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APPENDICES

Appendix A: Survey Instrument (Summary)

The survey instrument included items measuring digital connectivity intensity, after-hours engagement, multitasking, decision fatigue, decision quality, coping practices, and demographic characteristics. Responses were captured on a five-point Likert scale ranging from strongly disagree to strongly agree.

1. I frequently check or respond to work messages outside official working hours.

2. I receive work-related notifications frequently during the workday.

3. Digital communication platforms interrupt my focused work.

4. I feel expected to respond to work messages immediately.

5. I often multitask between multiple digital platforms during work.

6. Making multiple decisions throughout the day leaves me mentally exhausted.

7. I find it harder to concentrate on decisions later in the day.

8. Continuous work pressure reduces my mental clarity.

9. I believe constant connectivity affects the quality of managerial decisions.